

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**

**ASSESSMENT TOOL 1 – C Code Output Prediction**

|                         |                                                                                                                      |                      |                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------|
| <b>Course Code:</b>     | CSA0305                                                                                                              | <b>Course Title:</b> | Data Structures                              |
| <b>CO Assessed:</b>     | CO1 – Imitation (BL3, BL4)                                                                                           | <b>Assessment:</b>   | Assessment Tool 1 – C Code Output Prediction |
| <b>Weightage:</b>       | 35% of CIA                                                                                                           | <b>Total Marks:</b>  | 100                                          |
| <b>CO1 Statement:</b>   | <i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i> |                      |                                              |
| <b>Student Name:</b>    | M. VAISHNAV                                                                                                          | <b>Reg. No.:</b>     | 192524251                                    |
| <b>Section / Batch:</b> | 2025                                                                                                                 | <b>Date:</b>         | 10/07/2026                                   |

**Code of Conduct**

*I M. VAISHNAV (192524251) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student M. VAISHNAV (192524251)

**Instructions**

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20\*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic                                                                   | Focus                                      | Q Nos  | Marks            |
|-----------------------------------------------------------------------------------|--------------------------------------------|--------|------------------|
| Linked data structure                                                             | Basic data structure                       | 1 - 2  | 20               |
| Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure | Classifications of Linear data structure   | 3 - 4  | 20               |
| Search Data Structure                                                             | Linear Search and Binary Search            | 5 -6   | 20               |
| Recursion, Performance analysis                                                   | Recursion                                  | 7 - 8  | 20               |
| Array Data Structures                                                             | Implementations Using Array data structure | 9 - 10 | 20               |
| <b>GRAND TOTAL:</b>                                                               |                                            |        | <b>100 Marks</b> |

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## QUESTIONS

**Q1** ✓  
C Code – Array Traversal  
10 marks**Q2** ✓  
C Code – Linear Search  
10 marks**Q3** ✓  
C Code – Binary Search Mid  
10 marks**Q4** ✓  
C Code – Pointer with Array  
10 marks**Q5** ✓  
C Code – Array Update  
10 marks**Q6** ✓  
C Code – Linked List Node  
10 marks**Q1 C Code – Array Traversal**

10 marks

```
#include <stdio.h>
int main() {
    int arr[] = {5, 10, 15, 20};
    int i, sum = 0;
    for(i = 0; i < 4; i++)
        sum += arr[i];
    printf("%d", sum);
    return 0;
}
```

**Your Answer**

50

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C Code – Array Traversal  
10 marks**Q2** ✓  
C Code – Linear Search  
10 marks**Q3** ✓  
C Code – Binary Search Mid  
10 marks**Q4** ✓  
C Code – Pointer with Array  
10 marks**Q5** ✓  
C Code – Array Update  
10 marks**Q6** ✓  
C Code – Linked List Node  
10 marks**Q7** ✓**Q2 C Code – Linear Search**

10 marks

```
#include <stdio.h>

int main() {
    int arr[] = {2,4,6,8,10};
    int key = 8;
    int i;
    for(i=0; i<5; i++){
        if(arr[i] == key){
            printf("%d", i);
            break;
        }
    }
}
```

**Your Answer**

3

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## QUESTIONS

**Q1** ✓  
C Code – Array Traversal  
10 marks**Q2** ✓  
C Code – Linear Search  
10 marks**Q3** ✓  
C Code – Binary Search Mid  
10 marks**Q4** ✓  
C Code – Pointer with Array  
10 marks**Q5** ✓  
C Code – Array Update  
10 marks**Q6** ✓  
C Code – Linked List Node  
10 marks**Q3 C Code – Binary Search Mid**

10 marks

```
#include<stdio.h>
int main()
{
    int low=0,high=7;
    int mid;
    mid=(low+high)/2;
    printf("%d",mid);
}
```

Your Answer

3

 Save Answer

**CO1\_AT1\_C CODE OUTPUT PREDICTION**[← Back](#)**QUESTIONS****Q1**   
C Code – Array Traversal  
10 marks**Q2**   
C Code – Linear Search  
10 marks**Q3**   
C Code – Binary Search Mid  
10 marks**Q4**   
C Code – Pointer with Array  
10 marks**Q5**   
C Code – Array Update  
10 marks**Q6**   
C Code – Linked List Node  
10 marks**Q8 C Code - Nested Loop**

10 marks

```
for(j=0;j<n;j++)  
for(j=0;j<n;j++)  
printf("%d");
```

**Your Answer** $O(n^2)$  **Save Answer**

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## QUESTIONS

**Q1** ✓  
C Code - Array Traversal  
10 marks**Q2** ✓  
C Code - Linear Search  
10 marks**Q3** ✓  
C Code - Binary Search Mid  
10 marks**Q4** ✓  
C Code - Pointer with Array  
10 marks**Q5** ✓  
C Code - Array Update  
10 marks**Q6** ✓  
C Code - Linked List Node  
10 marks**Q9 C Code - Binary Search Complexity**

10 marks

```
while(low<=high){  
    mid=(low+high)/2;  
}
```

**Your Answer**

O(log n)

 **Save Answer**

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## QUESTIONS

**Q1** ✓  
C Code – Array Traversal  
10 marks**Q2** ✓  
C Code – Linear Search  
10 marks**Q3** ✓  
C Code – Binary Search Mid  
10 marks**Q4** ✓  
C Code – Pointer with Array  
10 marks**Q5** ✓  
C Code – Array Update  
10 marks**Q6** ✓  
C Code – Linked List Node  
10 marks**Q10 C Code - Pointer Increment**

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

## Your Answer

10

 Save Answer